

Decision-Calibrated Paired Evaluation with Sequentially Acquired Human Judgments

Anonymous Authors¹

Abstract

Pairwise language-model evaluations increasingly use inexpensive model judges, but deployment decisions concern a specified human population and must account for ties, prompt variation, correlated judge errors, and adaptively chosen human labels. We formulate decision-calibrated paired evaluation around the finite-frame estimand “human win probability minus loss probability” under fixed item, prompt, response-generation, rater, and display-order distributions. A tie-aware hierarchical jury model drives one-step value-of-information acquisition, while an explicit random-exploration mixture and logged draw probabilities preserve design-based identification. A prequential augmented Hansen–Hurwitz estimator is conditionally unbiased under stated response-stability, positivity, and boundedness conditions and supports fixed-budget martingale inference even when the jury model is misspecified. This methods-first draft specifies simulation, human-audit, and temporal-shift tests; efficiency and decision-quality claims remain pending empirical execution.

1. Introduction

Model comparison is often reduced to a leaderboard score, although the actual decision may be whether system A should replace system B for a particular population of requests. Human pairwise evaluation directly targets preference and underlies Chatbot Arena (Chiang et al., 2024), but comprehensive human labeling is expensive. Model-based judges scale cheaply (Zheng et al., 2023) yet exhibit position, style, verbosity, and self-preference effects (Panickssery et al., 2024; Ye et al., 2025). A jury of correlated judges cannot be treated as an equally large set of indepen-

dent humans.

Recent work estimates judge-specific bias and uncertainty (Dubois et al., 2026), shared confounding (Zhao et al., 2026), jury discrimination and internal consistency (Qian et al., 2026), and uncertainty with iterative comparison selection in comparative evaluation (Fathullah & Gales, 2025). Human-calibrated estimators correct noisy model judgments (Lee et al., 2026; Chen et al., 2026). Efficient inference for a static, independently selected calibration set is now well understood enough that “apply AIPW to judge labels” is not a new contribution.

The specific issue studied here is the combination of a tie-aware crossed human-preference frame, decision-loss acquisition, and sequential inference. Once human cells are chosen because previous outcomes made them decision-relevant, a final inclusion probability cannot generally be reconstructed by multiplying probabilities along the realized path. Ordinary post-hoc cross-fitting can leak future selection information, while pure value-of-information acquisition can assign zero probability to target cells.

We ask: *can adaptive human labeling reduce the cost of deciding whether A is materially better than B while retaining valid inference for an explicit human-preference estimand?* We make four scoped contributions:

- We define a tie-aware finite evaluation frame that fixes item, prompt, response coupling, rater, and display-order distributions before labeling.
- We use a crossed jury model only for prediction and decision-targeted acquisition, not as the definition of quality.
- We combine value-of-information acquisition with randomized exploration and a draw-level sequential augmented Hansen–Hurwitz estimator. We state its response-and-design assumptions and fixed-budget asymptotic conditions.
- We lay out planned simulations, a sealed target-random human audit, shift tests, and ablations that separate interval validity from label efficiency; executable DGPs remain a preregistration blocker.

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

Evidence status. The formal design claims are established below. Whether adaptive acquisition saves labels, whether the working model predicts humans, and whether the chosen human panel represents deployment stakeholders remain empirical questions; no such result is reported in this draft.

2. Related Work

Measurement variation and judge bias. Multi-prompt evaluation models performance distributions across prompt forms (Polo et al., 2024). Human studies can be assessed by the probability that a repeat study preserves a ranking (Riley et al., 2024). Model judges introduce additional structure: self-recognition can produce self-preference (Panickssery et al., 2024), controlled perturbations reveal multiple biases (Ye et al., 2025), and modern hierarchical methods estimate grader effects, confounding, or discrimination (Dubois et al., 2026; Zhao et al., 2026; Qian et al., 2026). Our jury model borrows this measurement perspective. Its novelty is not a new bias taxonomy or latent quality model.

Human-calibrated inference. Binary sensitivity/specificity correction with propagated calibration uncertainty and pilot-based allocation across true-label classes is studied by Lee et al. (2026). Chen et al. (2026) unify measurement-error and prediction-powered estimators for mean human outcomes, including pairwise win rates, under independently selected calibration/test designs. Those estimators do not by themselves justify reusing an outcome-adaptively acquired calibration sample. Classical unequal-probability with-replacement estimation (Hansen & Hurwitz, 1943) and inference for adaptively collected data (Hadad et al., 2021) supply the statistical foundation. Prediction-Powered Active Testing (PPAT) combines active-testing weights with a prediction control variate (Ashouritaklimi et al., 2026); Active Testing supplies the earlier adaptive risk estimation framework (Kossen et al., 2021), while control-variate evaluation combines synthetic and human feedback for unbiased head-to-head win rates (Zhou et al., 2025). Our distinction is decision-loss acquisition for three-way human comparisons over a crossed evaluation frame, not adaptive residualization or its central limit theorem by itself. DMLRank gives nonparametric, debiased inference for tie-aware preference rankings and derives budget-aware collection policies (Frauen et al., 2026). Its target is a multi-model ranking functional; ours is one frozen replacement decision for an explicitly weighted item–response–rater–order frame, with outcome-adaptive draw probabilities logged prequentially.

Adaptive and dynamic evaluation. Active acquisition can exploit historical dependence among benchmark items (Li et al., 2025), while ProEval uses surrogate models for performance estimation and failure discovery (Huang et al.,

2026). Such compression can fail under source–target model shift (Zhang et al., 2025). Dynamic benchmarks reduce static contamination through generated or refreshed tasks (Zhu et al., 2024; White et al., 2025); contamination tests can detect some forms of memorized ordering (Oren et al., 2024). Freshness, however, changes the evaluation distribution and complicates longitudinal claims. Finally, non-transitive judge preferences can make a scalar rank baseline-dependent (Xu et al., 2025). We therefore target one pre-specified pairwise replacement decision, not a universal leaderboard. SCOPE conformally calibrates abstention for pairwise model judgments under exchangeability (Badshah et al., 2026); its selective-judge risk is distinct from estimating a weighted human-panel preference with adaptive labels. Collaborative adaptive labeling also combines imperfect labels with selective expert escalation and dependence-aware estimation (Ruan et al., 2026). FairJudge instead trains the judge itself to reduce selected biases (Yang et al., 2026). Our model jury remains fallible: identification comes from randomized human acquisition, and the claimed novelty is limited to the tie-aware crossed decision frame and decision-loss policy. AURA adaptively prioritizes uncertain pairwise judge decisions for human verification (Zhang et al., 2026). It targets refinement of latent judge trust; our target is instead a prespecified weighted human-panel mean, with the acquisition model allowed to be wrong.

3. Evaluation Goal and Desiderata

At evaluation time t_0 , freeze models A, B , their decoding policies, the rubric, and a finite frame of N root items. Item i has target weight $w_i \geq 0$, $\sum_i w_i = 1$. Cell $c \in \mathcal{C}_i$ identifies a prompt variant and paired response realizations with prespecified conditional weight $a_{c|i}$. Independent response draws, shared random seeds, and deterministic decoding define different estimands because preference is nonlinear; the coupling is therefore fixed explicitly.

Before acquisition, enroll a finite rater panel $\mathcal{H} = \{1, \dots, H\}$ with nonnegative target weights ρ_h summing to one. Inference therefore targets this panel; extrapolation to other people is a separate superpopulation claim. For display order $o \in \{-1, +1\}$, define the outcome of one standardized rating draw

$$Y_{icho} = \begin{cases} +1, & A \text{ preferred,} \\ 0, & \text{tie,} \\ -1, & B \text{ preferred.} \end{cases} \quad (1)$$

and $\mu_{icho} = \mathbb{E}[Y_{icho} \mid i, c, h, o]$, where the expectation covers within-rater response variation under the fixed protocol. The finite-frame target is

$$\Delta_F = \sum_{i=1}^N w_i \sum_{c \in \mathcal{C}_i} a_{c|i} \sum_{h \in \mathcal{H}} \rho_h \frac{1}{2} \sum_{o \in \{-1, +1\}} \mu_{icho}. \quad (2)$$

Equivalently, define a target unit $x = (i, c, h, o) \in \mathcal{X}$ with mass

$$\pi(x) = w_i a_{c|i} \rho_h / 2, \quad \sum_{x \in \mathcal{X}} \pi(x) = 1. \quad (3)$$

Then $\Delta_F = P_\pi(A \text{ wins}) - P_\pi(B \text{ wins})$ and the half-tie win rate is $(1 + \Delta_F)/2$. We separately target

$$\tau_F = \sum_{x \in \mathcal{X}} \pi(x) P(Y_x = 0 \mid x) \quad (4)$$

because the same Δ_F can describe decisive disagreement or widespread ties.

A practical margin $\delta > 0$ and loss matrix are pre-specified. The action set is “ A materially better” if $\Delta_F > \delta$, “practically equivalent” if $|\Delta_F| \leq \delta$, “ B materially better” if $\Delta_F < -\delta$, and “not established” when data do not support a decisive action. This is an operational human-preference construct, not latent universal quality.

4. Evaluation Method

The protocol first collects inexpensive labels from a family-diverse model jury on all cells, then a uniformly randomized human pilot. Before each subsequent batch, a working model is trained only on prior batches and used to compute decision value. Human acquisition mixes this adaptive policy with the target item distribution. A sealed, uniformly sampled human audit is never used for fitting or acquisition. The implementation refits once per predeclared batch. It approximates each hypothetical EVI update by one frozen Laplace/Newton step, forms one p_b for the next batch, and logs that probability for every draw. The working Bayes-risk action used inside EVI is distinct from the terminal confidence-set action. Conditional on the batch-start history, draws are IID with replacement from this frozen law and the predictor is fixed; draw-level Equation (8) remains valid because both are predictable.

4.1. Data and Task Construction

The frame is a probability sample from a timestamped deployment log or a weighted benchmark with a clearly narrower finite-frame interpretation. Sampling excludes personal data, unsupported licenses, and items requiring professional expertise absent from the rater pool. Root-item provenance, inclusion weight, domain, language, safety category, near-duplicate cluster, prompt generator, response seeds, model revisions, tokenizer, and decoding parameters are versioned before labels are revealed. Each public root also records its pinned repository revision, export-byte hash, stable upstream identifier, canonical raw-row hash, and versioned normalized-content hash. Normalized duplicate clusters are formed before frame sampling and assigned as units;

any separately admitted pilot-fitting or sealed source is a distinct registered role, and cross-role clusters are quarantined. Paraphrase and template leakage remain separate blocking checks, since hash inequality is not evidence of semantic independence. A hash-bound preflight additionally rejects named PII/secret patterns and retained character-5-gram near-duplicates; it is a lexical heuristic, not a substitute for reviewed multilingual, paraphrase, privacy, or benchmark-leakage audits.

Each prompt variant is validated as meaning-preserving by a separate process; the target weights over variants are fixed, not chosen after seeing which model benefits. Model identifiers are hidden (although style may reveal identity), AB/BA display is randomized, and humans receive a real tie option with a fixed definition. “Equally good” and “both unacceptable” are recorded separately even if the primary analysis maps both to zero. Independent ratings define Equation (2); majority vote and adjudication are secondary, different estimands.

Rater and order assignments follow the target masses in Equation (3), or their logged adaptive counterparts. Availability is resolved before content is shown. Every assignment attempt is logged; a skip after viewing content is a separate outcome, never silently reassigned. The formal estimator below applies only to complete ratings. For non-response we report worst-case bounds and a separately labeled inverse-response analysis under a stated missing-at-random model; outcome-dependent skipping is not repaired by weighting alone.

4.2. Scoring and Aggregation

At round t , sample one complete target unit with replacement,

$$X_t = (I_t, C_t, H_t, O_t) \sim p_t(\cdot \mid \mathcal{F}_{t-1}), \quad (5)$$

where $\mathcal{F}_{t-1}$ contains every prior assignment, sampling probability, jury feature, outcome, and model random seed. This joint law makes rater and order assignment explicit. If operational constraints require a nested assignment, the product of logged item, cell, rater, and order probabilities must equal $p_t(X_t)$. Write Y_t^* for the standardized rating that would be recorded under a completed assignment. The design claim additionally requires stability of its full response law:

$$P(Y_t^* = y \mid \mathcal{F}_{t-1}, X_t = x) = p_y(x), \quad y \in \{-1, 0, 1\}, \quad (6)$$

where $\mu_x = \sum_y y p_y(x)$. Learning, fatigue, memory, or rubric drift can violate Equation (6). Randomized task order, training before study labels, and temporal diagnostics reduce but do not prove this assumption. The protocol randomizes attempt positions, interleaves hidden audit attempts, and retains repeated roots rather than resampling

them. A sensitivity analysis assumes the weaker uniform bound $|\mathbb{E}[Y_t | \mathcal{F}_{t-1}, X_t = x] - \mu_x| \leq \gamma$; then the design estimator’s absolute drift bias is at most γ , so both interval endpoints are expanded by the frozen $\gamma \in \{0, .02, .05\}$.

Let $m_{t-1}(x) \in [-1, 1]$ be a bounded prediction of Y_x trained only on prior rounds and frozen before draw t , and define

$$\bar{m}_{t-1} = \sum_{x \in \mathcal{X}} \pi(x) m_{t-1}(x). \quad (7)$$

The draw-level sequential augmented Hansen–Hurwitz contribution is

$$\psi_t = \bar{m}_{t-1} + \frac{\pi(X_t)}{p_t(X_t)} \{Y_{X_t} - m_{t-1}(X_t)\}, \quad (8)$$

with final estimator

$$\hat{\Delta}_T = \frac{1}{T} \sum_{t=1}^T \psi_t. \quad (9)$$

Repeated draws are retained. p_t is the conditional draw probability, not the probability that a unit is ever included; substituting a realized-path inclusion probability is generally invalid under outcome adaptation. Replacing Y_x in Equation (8) by $1\{Y_x = 0\}$ gives the tie-rate estimator, with its covariance with $\hat{\Delta}_T$ reported.

The fixed T counts assignment attempts, never replacement completions. Let R_t indicate observation and store $Y_t^{\text{obs}} = R_t Y_t^*$, where $Y_t^* \in [-1, 1]$; let ψ_t^* denote Equation (8) with this complete outcome. The primary report replaces a missing outcome pointwise by -1 and $+1$ in

$$\psi_t^\pm = \bar{m}_{t-1} + \frac{\pi(X_t)}{p_t(X_t)} [R_t \{Y_t^{\text{obs}} - m_{t-1}(X_t)\} + (1 - R_t) \{\pm 1 - m_{t-1}(X_t)\}]. \quad (10)$$

Thus $\psi_t^- \leq \psi_t^* \leq \psi_t^+$ for every assignment, without a missing-at-random assumption. Exact interval construction appears in Section 5.2. A separate analysis assumes $P(R_t = 1 | \mathcal{F}_{t-1}, X_t, Y_t^*) = e_t(X_t) \geq e_{\min} > 0$ and uses the predictable contribution

$$\psi_t^{\text{MAR}} = \bar{m}_{t-1} + \frac{\pi(X_t)}{p_t(X_t)} \frac{R_t}{e_t(X_t)} \{Y_t^{\text{obs}} - m_{t-1}(X_t)\}.$$

It is secondary because an estimated response model does not make outcome-dependent skipping ignorable.

4.3. Human or Model-Based Judging

The jury model is a prediction and diagnostic instrument, not the source of identification. A covariate-dependent gen-

eralization of the Davidson tie model (Davidson, 1970) uses

$$D_{jico} = e^{\eta_{jico}/2} + e^{-\eta_{jico}/2} + e^{\kappa_{jico}}, \quad (11)$$

$$P_j(Y = 1) = e^{\eta_{jico}/2} / D_{jico}, \quad (12)$$

$$P_j(Y = -1) = e^{-\eta_{jico}/2} / D_{jico}, \quad (13)$$

$$P_j(Y = 0) = e^{\kappa_{jico}} / D_{jico}. \quad (14)$$

One explicit working predictor is

$$\eta_{jico} = \alpha_j + d_j(\beta + b_i + b_{iv} + b_{ic}) + \gamma_j^O o + \gamma_j^L \ell_{ic} + \gamma_j^S s_{jic} + u_{j,d(i)}, \quad (15)$$

$$\kappa_{jico} = \kappa_j + t_i + t_{iv} + z_{ic}^\top \xi_j, \quad (16)$$

with $d_j > 0$ and partially pooled effects. Identifiability constraints set weighted human-panel offset to zero and weighted log-discrimination to zero, center item and variant effects, and require a connected model-judge–cell overlap graph with both orders and estimable covariate contrasts. The public pilot has one checkpoint per jury family, so it does not estimate a separate family-variance component. Human-rater effects are partially pooled under a proper prior and do not require every rater to appear in the 200-attempt pilot. Conditional independence is only a working assumption. The two recorded tie subtypes are collapsed to zero before this three-category model and analyzed separately.

The human scale is anchored by the frozen weighted panel rather than assuming jury consistency equals validity. Length and family coefficients remain descriptive unless randomized interventions identify them: removing a length association can remove genuine quality as well as bias.

Decision-targeted acquisition. The protocol fixes the normalized loss matrix below, with rows (A, B, E, NE) and true states (A, B, E) :

$$L = \begin{pmatrix} 0 & 4 & 2 \\ 4 & 0 & 2 \\ 2 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}. \quad (17)$$

Stakeholder-specific alternatives are sensitivity analyses, not post-hoc replacements. Let ϑ collect the finite working-model coefficients. At batch start b , freeze the Laplace posterior $q_b(\vartheta) = \mathcal{N}(\hat{\vartheta}_b, H_b^{-1})$, the deterministic frame functional $\Delta(\vartheta)$, and

$$Q_b(S = s) = \int 1\{S(\Delta(\vartheta)) = s\} q_b(\vartheta) d\vartheta, \quad (18)$$

$$\hat{p}_b(y | x) = \int p_\vartheta(y | x) q_b(\vartheta) d\vartheta. \quad (19)$$

Both integrals use frozen common Monte Carlo draws. Define Bayes risk

$$\tilde{R}(Q) = \min_{d \in \{A, B, E, \text{NE}\}} \sum_s L_{ds} Q(S = s), \quad (20)$$

breaking action ties in the fixed order NE, E , A , B . Let $q_b^{x,y}$ be obtained from q_b by one predeclared damped Newton update of the mode and observed Hessian using $\log p_\vartheta(y | x)$, with frozen prior, ridge, damping, and no-line-search rule. Then $U(Q_b; x, y)$ is the state distribution induced by $q_b^{x,y}$ through the same $\Delta(\vartheta)$ and common draws. Neither the update nor its hyperparameters depend on p_b . The update evaluates gradient and Hessian at $\hat{\vartheta}_b$, uses the frozen ridge/damping formula in the preregistration, and factorizes with a fixed jitter ladder. Standard-normal draws use seed 2026090322. If every registered Cholesky attempt fails, that cell receives EVI zero and the failure is retained rather than silently changing the approximation. For target unit x with annotation cost k_x , define

$$\text{EVI}_b(x) = \tilde{R}(Q_b) - \sum_{y \in \{-1, 0, 1\}} \hat{p}_b(y | x) \tilde{R}\{U(Q_b; x, y)\}. \quad (21)$$

Using predeclared temperature λ_E , define

$$g_b(x) = \lambda_E \text{EVI}_b(x) / k_x, \quad (22)$$

$$r_b(x) = \exp\{g_b(x)\} / \sum_{x'} \exp\{g_b(x')\}, \quad (23)$$

$$p_b(x) = \epsilon \pi(x) + (1 - \epsilon) r_b(x), \quad \epsilon > 0. \quad (24)$$

All EVI values are computed before p_b , after which Q_b, p_b, m_b , and $\bar{m}_b$ are frozen for the complete batch; each draw-level p_t equals that p_b . This removes any dependence of EVI on its own acquisition probability and gives $p_t(x) \geq \epsilon \pi(x)$ and $\pi(x) / p_t(x) \leq 1/\epsilon$. Exact EVI is myopically optimal only under a correct posterior, exact update, and no exploration constraint; the implemented approximation claims neither finite-horizon optimality nor a general regret bound. It scores approximate Bayesian risk reduction, whereas the terminal action uses a confidence set; their alignment is a required simulation diagnostic, not an identity.

5. Validation

Proposition 5.1 (Sequential unbiasedness). *Assume the finite frame and fixed budget T are prespecified; Equation (6) holds; outcomes are complete; p_t and m_{t-1} are measurable before outcome t ; sampling is with replacement from the logged p_t ; and Equation (24) holds. Then*

$$\mathbb{E}[\psi_t | \mathcal{F}_{t-1}] = \Delta_F, \quad \mathbb{E}[\hat{\Delta}_T] = \Delta_F, \quad (25)$$

for any misspecified working jury model or outcome predictor.

The proof follows by conditioning on X_t and appears in Section A. Forward or prequential fitting is essential: each

prediction and sampling probability is frozen before its outcome. Ordinary random K -fold fitting after adaptive collection can leak information through future selections.

Theorem 5.2 (Fixed-budget inference). *Under the assumptions of Proposition 5.1, let $D_{Tt} = \psi_{Tt} - \Delta_F$. Suppose*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[D_{Tt}^2 | \mathcal{F}_{t-1}] \xrightarrow{p} V \in (0, \infty) \quad (26)$$

for deterministic V , and for every $\eta > 0$ the martingale Lindeberg quantity

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[D_{Tt}^2 \mathbf{1}\{|D_{Tt}| > \eta\sqrt{T}\} | \mathcal{F}_{t-1}] \xrightarrow{p} 0. \quad (27)$$

Then

$$\sqrt{T}(\hat{\Delta}_T - \Delta_F) \xrightarrow{d} \mathcal{N}(0, V). \quad (28)$$

This is the standard martingale central-limit route (Hall & Heyde, 1980), specialized to predictable adaptive sampling. Because $Y, m \in [-1, 1]$ and $\pi/p \leq 1/\epsilon$, increments are bounded when ϵ is fixed. This boundedness gives the required squared-increment martingale law of large numbers, so

$$\hat{V}_T = \frac{1}{T-1} \sum_{t=1}^T (\psi_t - \hat{\Delta}_T)^2, \quad \hat{\text{se}} = \sqrt{\hat{V}_T / T} \quad (29)$$

gives studentized fixed-budget inference. This is pointwise asymptotic, not exact finite-sample, coverage. The protocol fixes T before the first human outcome and forbids threshold-based optional stopping.

5.1. Contamination and Gaming Resistance

Public static items are checked for exact, normalized, paraphrase, and template-level overlap with model-accessible corpora where those corpora are available. Ordering-based tests provide evidence for only particular forms of memorization (Oren et al., 2024); absence of a signal is not proof of cleanliness. A fresh private wave and generated variants reduce exposure but create a new population, so longitudinal analyses retain bridge items and separate within-wave model difference from between-wave drift.

Judge prompts and identities are hidden from evaluated systems. Position is randomized, response identifiers are stripped, and controlled style/length transformations probe judge sensitivity. Model outputs containing instructions to the evaluator are separately flagged. The human-only sealed audit remains the final check against jury manipulation.

5.2. Statistical Protocol

When every assignment is complete, the primary two-sided interval uses the frozen $z_{0.975} = 1.959963984540054$:

$$[L_T^{\text{cmp}}, U_T^{\text{cmp}}] = \left[\hat{\Delta}_T \pm z_{0.975} \sqrt{\hat{V}_T/T} \right] \cap [-1, 1]. \quad (30)$$

If any $R_t = 0$, define $B_\epsilon = 1 + 2/\epsilon$, $c_\alpha = B_\epsilon \sqrt{2 \log(2/\alpha)/T}$, and use instead

$$\begin{aligned} L_T^{\text{miss}} &= \max \left\{ -1, \frac{1}{T} \sum_t \psi_t^- - c_\alpha \right\}, \\ U_T^{\text{miss}} &= \min \left\{ 1, \frac{1}{T} \sum_t \psi_t^+ + c_\alpha \right\}. \end{aligned} \quad (31)$$

Each endpoint contribution lies in $[-B_\epsilon, B_\epsilon]$ and its predictable mean bounds Δ_F ; two one-sided martingale–Hoeffding bounds give coverage at least $1 - \alpha$ without a response-mechanism assumption. Write $[L_T, U_T]$ for the applicable interval. It is widened by γ in the response-drift sensitivity analysis. This guarantee is intentionally severe: at the pilot values $\epsilon = 0.2$, $T = 1000$, and $\alpha = 0.05$, $c_\alpha = 0.9448$. Consequently even one missing outcome will generally force NE; simulation must report this degeneracy rather than presenting the bound as label-efficient. The terminal action used with Equation (17) and in the final report is

$$\hat{a} = \begin{cases} A, & L_T > \delta, \\ B, & U_T < -\delta, \\ E, & [L_T, U_T] \subseteq [-\delta, \delta], \\ \text{NE}, & \text{otherwise.} \end{cases} \quad (32)$$

For complete outcomes this wrong-action probability is at most $\alpha + o(1)$; with Equation (31) it is at most α under response stability, for this single comparison. This does not bound abstention, expected stakeholder loss, or multi-model ranking error.

Theorem 5.2 is conditional on the full realized frame, including the enrolled rater panel and fixed response pairs. For a separate deployment-population target Δ_{pop} ,

$$\hat{\Delta}_T - \Delta_{\text{pop}} = (\hat{\Delta}_T - \Delta_F) + (\Delta_F - \Delta_{\text{pop}}). \quad (33)$$

The martingale variance covers only the first term. The second requires the actual first-stage item and rater recruitment design and its replicate weights or a justified superpopulation model; prompt variants cannot be bootstrapped as independent root items.

6. Empirical Study

Simulation precedes human collection. We vary Δ_F at $\{-\delta, 0, \delta\}$ and local alternatives $\delta \pm c/\sqrt{T}$, tie prevalence,

item correlation, judge-family correlation, repeated-rater effects, order/length/family effects, outcome-model misspecification, rare strata, small propensities, nonresponse, and temporal shift. For each condition, 2,000 Monte Carlo trials compare target-random human sampling, static independent calibration, Active Testing, control-variate evaluation, PPAT-style residual acquisition, variance-optimal residual sampling, pure uncertainty sampling, and EVI. Each acquisition policy is crossed with the same valid Equation (9); augmentation is then ablated at fixed policy. Primary metrics are bias, RMSE, 95% coverage, interval width, false decisive actions, equivalence decisions, abstention, realized loss, maximum weight, and ESS, with Monte Carlo uncertainty. The simulation separately reports whether Bayes-risk acquisition actually lowers realized loss under the terminal confidence-set action.

The planned public study freezes all 200 roots at the pinned LiveBench reasoning revision, 200 hash-ranked roots from math, and all 128 coding roots: a 528-root weighted public-benchmark frame. Each source receives mass 1/3 and roots are uniform within source. This is neither a product-domain sample nor a deployment population. It uses two validated prompt forms per item and paired response draws from two fixed open model revisions. Four model-judge families score all cells in both orders. A 200-rating uniform human pilot precedes 800 sequential ratings; a separate 200-rating target-random audit sampled from Equation (3) remains sealed. All counts and fixed budget $T = 1000$ assignment attempts are set by simulation before the first human outcome; a missing response is not replaced. A frozen panel of 80 raters is recruited against a written definition, assigned target weights, trained on a fixed handbook, and screened with non-study examples. The expected load is 12.5 primary attempts per rater; realized load, withdrawals, and response are reported. All T assignment slots are preassigned. Rater unavailability, withdrawal, or a binding quota records $R_t = 0$ in its original slot; no replacement occurs, later slots continue, and the analysis remains at fixed T using the predeclared partial-identification bounds. Identifiers are hidden, both display orders are balanced in expectation, and every skip or nonresponse is retained. Attempt position is randomized, sealed audit attempts are interleaved at hidden prespecified slots, and the drift sensitivity grid is frozen before labels.

6.1. Planned Primary Endpoints

Table 1. Primary finite-frame decision results. Dashes indicate that human collection has not been run. The table will report tie rate and interval, not a leaderboard rank.

Design	n_H	$\hat{\Delta}_F$ [95% CI]	Tie	Action
Target random	–	–	–	pending
Static calibration	–	–	–	pending
PPAT-style	–	–	–	pending
EVI	–	–	–	pending
Sealed audit	–	–	–	pending

Efficiency will be claimed only if interval validity is retained in simulation and an interval for the adaptive-minus-audit difference lies within a frozen equivalence margin using a frozen joint-covariance procedure. Neither is yet registered, so the current draft permits no equivalence claim. Jury-only rankings are diagnostic and never substitute for Equation (2).

6.2. Robustness and Ablations

We ablate augmentation, EVI, the exploration floor, tie modeling, root-item effects, judge-family effects, and prequential fitting while crossing each acquisition policy with a common estimator. Acquisition trained on older model pairs is tested on a held-out stronger pair to expose frontier shift. Additional falsification tests swap AB/BA order, control response length and style, hold out prompt families, remove one judge family, change the practical margin, and repeat the study on a fresh temporal wave.

Model checks use held-out log score, multiclass Brier score, tie calibration, swap symmetry, posterior predictive checks, and residuals by domain, order, length, family, judge, and item. Disagreement cases are sampled by a prespecified stratified rule rather than selected for narrative convenience. SCOPE-style bidirectional conformal abstention is included as a jury-only diagnostic and never substitutes for the human estimand.

7. Limitations

The finite frame may omit important deployment requests, and a convenience rater panel supports inference only to that panel. Human preference is not factual correctness, safety, fairness, or welfare. The Davidson model can be misspecified, judge correlations can be more complex than modeled, and non-transitivity can make broader rankings incoherent. Sequential unbiasedness does not guarantee useful finite-sample intervals when exploration probabilities are small.

The method requires exact logging of every adaptive probability and a fixed budget or sequentially valid interval. EVI

may save no labels and can be worse than uniform sampling under model shift. A sealed audit consumes additional human effort. Fresh items reduce exposure but impair longitudinal comparability, while black-box contamination cannot be ruled out. Finally, one pairwise replacement decision does not establish a general benchmark or model ordering.

8. Conclusion

We framed language-model comparison as a decision about a frozen, weighted human panel rather than a latent jury score. Randomized exploration and draw-level sequential augmentation permit outcome-adaptive human acquisition without surrendering design identification under explicit assumptions. The formal result does not show that acquisition is efficient or that the working model is valid. The decisive next evidence is coverage under misspecified, correlated simulations followed by agreement with a sealed uniform human audit.

Impact Statement

Evaluation methods shape which systems are deployed and which capabilities are optimized. A narrow or unrepresentative item and rater frame can legitimize a harmful deployment even when its confidence interval is statistically valid. Adaptive labeling may undersample rare communities unless target weights and the exploration floor preserve them. Raters may encounter disturbing content; they require informed consent, task-specific warnings, the ability to skip without penalty, fair compensation, and access to support. Deployment logs must be de-identified and retained under strict access controls. Public reporting will include the population definition, tie rate, abstentions, maximum sampling weights, subgroup support, and unresolved validity threats. The method is not intended to turn human preference into a universal quality score or to justify deployment without separate safety and factuality audits.

References

- Ashouritaklimi, K., Kilian, V., Huang, D., Rainforth, T., and Caron, F. Prediction-powered active testing, 2026. URL <https://arxiv.org/abs/2607.08347>.
- Badshah, S., Emami, A., and Sajjad, H. SCOPE: Selective conformal optimized pairwise LLM judging. In *International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/62431>.
- Chen, Y. T., Lu, S., Li, S., Guo, M., and Li, S. Efficient inference for noisy LLM-as-a-judge evaluation. In *International Conference on Machine Learning*, 2026. URL

- <https://icml.cc/virtual/2026/poster/61131>.
- Chiang, W.-L., Zheng, L., Sheng, Y., Angelopoulos, A. N., Li, T., Li, D., Zhu, B., Zhang, H., Jordan, M., Gonzalez, J. E., and Stoica, I. Chatbot Arena: An open platform for evaluating LLMs by human preference. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 8359–8388. PMLR, 2024. URL <https://proceedings.mlr.press/v235/chiang24b.html>.
- Davidson, R. R. On extending the Bradley–Terry model to accommodate ties in paired comparison experiments. *Journal of the American Statistical Association*, 65(329): 317–328, 1970. doi: 10.1080/01621459.1970.10481082.
- Dubois, M., Coppock, H., Giulianelli, M., Jorgensen, O., Flesch, T., Luettgau, L., and Ududec, C. Quantifying biases in LLM-as-a-judge evaluations. In *International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/61443>.
- Fathullah, Y. and Gales, M. Generalised probabilistic modelling and improved uncertainty estimation in comparative LLM-as-a-judge. In *Proceedings of the Forty-First Conference on Uncertainty in Artificial Intelligence*, volume 286 of *Proceedings of Machine Learning Research*, pp. 1266–1288. PMLR, 2025. URL <https://proceedings.mlr.press/v286/fathullah25a.html>.
- Frauen, D., Deviyani, A., van der Schaar, M., and Feuerriegel, S. Nonparametric LLM evaluation from preference data. In *International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/61327>.
- Hadad, V., Hirshberg, D. A., Zhan, R., Wager, S., and Athey, S. Confidence intervals for policy evaluation in adaptive experiments. *Proceedings of the National Academy of Sciences*, 118(15):e2014602118, 2021. doi: 10.1073/pnas.2014602118.
- Hall, P. and Heyde, C. C. *Martingale Limit Theory and Its Application*. Academic Press, 1980.
- Hansen, M. H. and Hurwitz, W. N. On the theory of sampling from finite populations. *The Annals of Mathematical Statistics*, 14(4):333–362, 1943. doi: 10.1214/aoms/1177731356.
- Huang, Y., Zeng, W., Kumaresan, A., and Wang, Z. ProEval: Proactive failure discovery and efficient performance estimation for generative AI evaluation. In *International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/62850>.
- Kossen, J., Farquhar, S., Gal, Y., and Rainforth, T. Active testing: Sample-efficient model evaluation. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 5753–5763. PMLR, 2021. URL <https://proceedings.mlr.press/v139/kossen21a.html>.
- Lee, C., Zeng, T., Jeong, J., Sohn, J.-y., and Lee, K. How to correctly report LLM-as-a-judge evaluations. In *International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/63607>.
- Li, Y., Ma, J., Ballesteros, M., Benajiba, Y., and Horwood, G. Active evaluation acquisition for efficient LLM benchmarking. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 35581–35602. PMLR, 2025. URL <https://proceedings.mlr.press/v267/li25bp.html>.
- Oren, Y., Meister, N., Chatterji, N., Ladhak, F., and Hashimoto, T. B. Proving test set contamination in black-box language models. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/46e624c244cff669223d488defd4e835-Abstract-Conference.html.
- Panickssery, A., Bowman, S. R., and Feng, S. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems*, volume 37, pp. 68772–68802, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/7f1f0218e45f5414c79c0679633e47bc-Abstract-Conference.html.
- Polo, F. M., Xu, R., Weber, L., Silva, M., Bhardwaj, O., Choshen, L., de Oliveira, A. F. M., Sun, Y., and Yurochkin, M. Efficient multi-prompt evaluation of LLMs. In *Advances in Neural Information Processing Systems*, volume 37, pp. 22483–22512, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/28236482f64a72eec43706b6f3a6c511-Abstract-Conference.html.
- Qian, M., Sun, G., Gales, M. J. F., and Knill, K. M. Who can we trust? LLM-as-a-jury for comparative assessment. In *International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/64228>.
- Riley, P., Deutsch, D., Foster, G., Ratnakar, V., Dabirmoghaddam, A., and Freitag, M. Finding replicable

- human evaluations via stable ranking probability. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 4908–4919. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.naacl-long.275. URL <https://aclanthology.org/2024.naacl-long.275/>.
- Ruan, X., Jia, N., Wei, W., Huang, S., Zhao, Z., Zheng, Z., and Wang, J. Collaborative adaptive labeling with imperfect labelers and selective expert escalation. In *ICML Workshop on Combining Theory and Benchmarks*, 2026. URL <https://icml.cc/virtual/2026/68297>.
- White, C., Dooley, S., Roberts, M., Pal, A., Feuer, B., Jain, S., Shwartz-Ziv, R., Jain, N., Saifullah, K., Dey, S., Shubh-Agrawal, Sandha, S. S., Naidu, S. V., Hegde, C., LeCun, Y., Goldstein, T., Neiswanger, W., and Goldblum, M. LiveBench: A challenging, contamination-limited LLM benchmark. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/e4a46394ba5378b3f9a186a5b4c650d1-Abstract-Conference.html.
- Xu, Y., Ruis, L., Rocktäschel, T., and Kirk, R. Investigating non-transitivity in LLM-as-a-judge. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 69583–69612. PMLR, 2025. URL <https://proceedings.mlr.press/v267/xu25w.html>.
- Yang, B., Feng, L., Chen, Y., Xu, X., Zhang, Y., and Li, S. FairJudge: An adaptive, debiased, and consistent LLM-as-a-judge. In *International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/64266>.
- Ye, J., Wang, Y., Huang, Y., Chen, D., Zhang, Q., Moniz, N., Gao, T., Geyer, W., Huang, C., Chen, P.-Y., Chawla, N. V., and Zhang, X. Justice or prejudice? quantifying biases in LLM-as-a-judge. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/fdca08d371e4b6c031397909e20043bd-Abstract-Conference.html.
- Zhang, G., Dörner, F. E., and Hardt, M. How benchmark prediction from fewer data misses the mark. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/c57bacb5ea621582235c54ce2a83136d-Abstract-Conference.html.
- Zhang, Z., Hung, Y.-T., He, W., Zhang, J., Ding, L., and Yeh, C.-K. AURA: Adaptive uncertainty-aware refinement for LLM-as-a-judge auditing, 2026. URL <https://arxiv.org/abs/2606.19714>.
- Zhao, J., Shin, C., Huang, T.-H., GNVV, S. S. S. N., and Sala, F. CARE: Confounder-aware aggregation for reliable LLM evaluation. In *International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/66456>.
- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., and Stoica, I. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36, pp. 46595–46623, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html. Datasets and Benchmarks Track.
- Zhou, Z., Song, Y., and Zanette, A. Accelerating unbiased LLM evaluation via synthetic feedback. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 79150–79167. PMLR, 2025. URL <https://proceedings.mlr.press/v267/zhou25v.html>.
- Zhu, K., Chen, J., Wang, J., Gong, N. Z., Yang, D., and Xie, X. DyVal: Dynamic evaluation of large language models for reasoning tasks. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/4ea4a1ea4d9ff273688c8e92bd087112-Abstract-Conference.html.

A. Datasheets and Reproducibility Details

A.1. Required Evaluation Manifest

The artifact records the frame timestamp, root-item source and target weight, deduplication cluster, domain and language, every prompt variant and weight, response coupling and seed, model and tokenizer revisions, decoder parameters, judge model and prompt revisions, order randomization, rater eligibility and training, rubric version, tie subtype, annotation cost, all $p_t(x)$ values before outcome reveal, outcome-model revision, EVI values, stopping rule, and exclusion reason. Probability normalization and positivity are checked online; a failed check halts collection.

The repository runner `python3 scripts/run_experiment_pipeline.py --plan experiments/eval_pipeline.json` enumerates frame acquisition, deduplication and response generation, jury-model construction, simulation and adaptive acquisition, and sealed evaluation. The committed JSON is a manifest validator and stage contract tied to the hashed evaluation section of `experiments/pilot_preregistration.json`; site-local commands are required for execution, after which output hashes and costs are recorded. The public-source registry pins each pilot frame and model revision. Automated license metadata is only a preflight; acquisition remains blocked until manual provenance, terms, privacy, human-subjects, and leakage reviews are approved. After those approvals, `scripts/materialize_public_data.py` verifies hash-pinned local exports, writes row-level provenance, and rejects retained content overlap across declared roles. Its only clearance bypass is a committed synthetic collision fixture whose artifacts are explicitly non-empirical. `scripts/run_content_gates.py` binds the frozen `experiments/content_gates.json` contract to those hashes and blocks retained named PII/secret patterns and lexical near-duplicates. Its large-artifact MinHash-LSH candidate generation can miss qualifying pairs, so that path is always inconclusive and blocking pending a reviewed recall-complete audit. An exhaustive pass still cannot establish semantic independence or approve a pending source review. The same section fixes the Qwen, SmoLLM, Mistral, and Phi jury revisions, posterior prior scales, one-step damped Newton approximation, unit annotation cost, nonresponse rule, and planned simulation factor levels. These values specify a public pre-pilot; they do not replace a deployment-log sampling design. The prompt-generator and rendered model prompts, variant weights, decoder and response-coupling seeds, response artifacts, and executable simulation DGPs are still unfrozen. Real response generation or human acquisition is blocked until a replacement preregistration hashes those estimand-defining objects. The companion `synthetic_commands.json` fixes a 16-unit frame, $\epsilon = 0.2$, $T = 300, 2,000$ trials, and seed 20260902. It uses deliberately misspecified prequential predictions to check the exact one-draw identity, positivity and weight bounds, fixed-budget studentization, and the tie estimator. It assumes complete response and is only a software audit; it does not replace the preregistered condition grid or provide human evidence.

A.2. Proof of Proposition 5.1

Conditioning on pre-outcome history,

$$\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = \sum_x p_t(x) \left[\bar{m}_{t-1} + \frac{\pi(x)}{p_t(x)} \{\mu_x - m_{t-1}(x)\} \right] \quad (34)$$

$$= \bar{m}_{t-1} + \sum_x \pi(x) \{\mu_x - m_{t-1}(x)\} = \Delta_F. \quad (35)$$

Averaging over fixed T proves the second statement. Neither step requires a correct jury model.

A.3. Consistency of the Variance Estimate

Let $D_{Tt} = \psi_{Tt} - \Delta_F$. Fixed $\epsilon > 0$ makes D_{Tt} uniformly bounded. Hence $D_{Tt}^2 - \mathbb{E}[D_{Tt}^2 \mid \mathcal{F}_{t-1}]$ is a bounded martingale difference and its average converges to zero in probability. The conditional variance assumption in Theorem 5.2 then yields $T^{-1} \sum_t D_{Tt}^2 \rightarrow_p V$; replacing Δ_F by $\hat{\Delta}_T$ changes the centered second moment by $o_p(1)$. Therefore $\hat{V}_T \rightarrow_p V$ and Slutsky's theorem justifies the stated studentized interval.

A.4. Human Audit

The audit draws complete (i, c, h, o) units from π before adaptation and is stored behind a separate random seed. It estimates the same finite-frame target from 200 fixed audit slots by the unweighted sample mean. Missing audit outcomes are replaced by -1 and $+1$ for lower and upper endpoint means, with one-sided Hoeffding radius $\sqrt{2 \log(2/\alpha)/200}$; they are never

dropped or replaced. If operational assignment changes the draw law, the exact target-to-draw ratio is retained and the corresponding bounded-weight interval is used. Audit membership never removes a unit from adaptive support: only the audit seed, assignments, and outcomes are hidden. Assignment seeds are independent, but outcomes from shared raters may be correlated and are not treated as independent in a future contrast. Its outcomes are opened only after the adaptive algorithm, model hyperparameters, and primary analysis are frozen. An adaptive-minus-audit equivalence test remains blocked until its margin and shared-rater covariance procedure are frozen.

B. Additional Validity Evidence

Table 2. Mandatory validity report; all values are pending collection.

Diagnostic	Value
Simulation coverage at $\Delta_F = \delta$	–
False decisive-action rate	–
Maximum $\pi(X_t)/p_t(X_t)$	–
Adaptive-sample ESS	–
Human inter-rater agreement	–
Jury multiclass Brier score	–
AB/BA swap discrepancy	–
Adaptive–audit estimate difference	–
Temporal-wave drift	–

Full per-domain estimates, tie subtypes, disagreement matrices, acquisition trajectories, all simulation conditions, and randomly selected qualitative cases will be reported. Post-hoc analyses are labeled exploratory and cannot replace the endpoints in Table 1.